Group 1 – Candlestick Image-Based Stock Trend Prediction using ViT & XAI

Objective:

The goal of this assignment is to develop a deep learning framework for stock trend prediction by transforming financial time-series data into candlestick chart images and solving the problem as an image classification task. Students will use a Vision Transformer (ViT) and apply Explainable Artificial Intelligence (XAI) techniques to interpret learned patterns.

Task Breakdown:

1. Dataset Collection & Preprocessing:

- Collect historical intraday stock market data (1-hour interval).
- Convert time-series data into candlestick chart images.
- Ensure uniform image size and format.
- Split dataset into Training (70%), Validation (20%), Testing (10%).

2. Labeling Strategy:

- Label images based on next day closing price:
  - Up
  - Down
  - Neutral
- Clearly define threshold.

3. Dataset Preparation:

- Organize dataset into train/validation/test folders.
- Ensure proper mapping between images and labels.

4. Model Implementation:

- Implement Vision Transformer (ViT).
- Use pretrained weights if available.
- Train using PyTorch/TensorFlow.
- Tune hyperparameters.

5. Explainability (XAI):

- Apply attention visualization / Grad-CAM.
- Analyze important regions in candlestick charts.

6. Evaluation & Analysis:

- Evaluate using Accuracy, Precision, Recall, F1-score.
- Visualize predictions.
- Analyze learned patterns.

7. Comparative Study:

- Compare with traditional patterns (Doji, Hammer, Engulfing).
- Discuss similarities/differences.

8. Final Report & Observations:

- Document findings.
- Discuss effectiveness, limitations, and future scope.

Deliverables:

- Code Implementation
- Public GitHub Repository (mandatory)
- Report (PDF)
- Visualization outputs

Discussion:

Discuss whether deep learning can rediscover classical candlestick patterns and identify new predictive structures.